

Problem 3.16

Suppose that $i_n = 5\%$ for all $n \geq 1$. How long will it take an investment to triple in value?

Solution

Since the effective annual interest rate is constant at 5%, the accumulation after n years is

$$P(1+i)^n.$$

To triple in value, we need

$$3P = P(1+i)^n.$$

With $i = 0.05$, this becomes

$$3P = P(1.05)^n.$$

Dividing by P ,

$$3 = (1.05)^n.$$

Taking natural logarithms,

$$\ln(3) = n \ln(1.05).$$

Therefore,

$$n = \frac{\ln(3)}{\ln(1.05)}.$$

Hence,

$$n \approx 22.517085.$$